

JACKSON ESHBAUGH

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Bethlehem, PA — 18020, USA

EDUCATION

• Lafayette College

B.S., Computer Science

◦ GPA: 4.00/4.00

August 2023–May 2027

Easton, PA

• Lafayette College

A.B., French Language

◦ GPA: 4.00/4.00

August 2023–May 2027

Easton, PA

RESEARCH INTERESTS

Topics

Computational Linguistics Machine Learning Interpretability
Theoretical Foundations of Artificial Intelligence

Guiding Questions

How do linguistic structure and semantics interact in multilingual NLP models?

How does language work—in computers and in humans?

How do we make neural networks more transparent?

PUBLICATIONS & PREPRINTS

C=CONFERENCE, W=WORKSHOP, J=JOURNAL, P=PREPRINT, S=IN SUBMISSION, T=THESIS

- S.1** Eshbaugh, J., Tiwari, C., and Silveyra, J. *A Modular and Multimodal Generative AI Framework for Urban Building Energy Data: Generating Synthetic Homes*. 2026. Submitted to *Building Simulation*. arXiv:2509.09794.
- W.1** Eshbaugh, J. *“Faithful to What?” On the Limits of Fidelity-Based Explanations*. 2026. *Workshop on Scientific Methods for Understanding Deep Learning at the International Conference on Learning Representations*. arXiv:2506.12176.

POSTERS & PRESENTATIONS

P=POSTER, T=TALK, K=KEYNOTE

- P.3** Eshbaugh, J. (Dec 2025). “Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data.” *Lafayette College EXCEL Scholars Poster Session*, Easton, PA. [poster]
- P.2** Eshbaugh, J. (Oct 2025). “Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data.” *CyberAccelerate Poster Session at KINBERCON 2025*, Lancaster, PA. [poster]
- P.1** Eshbaugh, J. (Sep 2025). “Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data.” *Lafayette College Bicentennial Weekend Poster Session*, Easton, PA. [poster]

Upcoming & Invited Presentations

- T.1** Silveyra, J. and Eshbaugh, J. (March 2026). “Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data.” *Invited talk at Georgia State University*, Atlanta, GA, USA.
- P.4** Eshbaugh, J. (April 2026). “Faithful to What? On the Limits of Fidelity-Based Explanations.” *Upcoming presentation at the Workshop on Scientific Methods for Understanding Deep Learning, ICLR 2026*, Rio de Janeiro, Brazil.

RESEARCH EXPERIENCE

• Independent Study—Linguistics of Idioms

Department of Language and Literary Studies, Lafayette College

◦ Advisor: Dr. Maria Hernandez

◦ Studying foundational linguistic theory, including de Saussure, to understand how idiomatic language functions within formal linguistic frameworks

◦ Examining implications for how idioms are theorized and processed

◦ Crafting a journal article in French that shares findings and implications

January 2026 - May 2026

Easton, PA

• Honors Thesis Research—Computational Linguistics

Department of Computer Science, Lafayette College

February 2025 - Present

Easton, PA

- Advisor: Dr. Sofia Serrano (Computer Science)
- Formalizing prior linguistic work on idiomaticity into a computational framework for model design
- Developing FRIdiom: an annotated corpus for detecting idiomatic expressions in French
- Applying back-translation and neural methods to improve figurative language handling in multilingual MT
- Using interpretability techniques from prior work to probe language model representations
- Connecting linguistic and computational methods to determine how humans and computers process idiomatic language
- Dataset paper in preparation for submission
- **Independent Research—Neural Network Interpretability** April 2025 - Present
Department of Computer Science, Lafayette College Easton, PA
 - Investigated when linear models fail to faithfully represent neural networks
 - Designed experimental framework comparing surrogate fidelity vs. task accuracy across regression tasks
 - Proposed the λ -score as a diagnostic metric; demonstrated high surrogate fidelity does not imply accuracy
 - Implemented full ML pipeline: network training, surrogate evaluation, statistical analysis
 - Manuscript reviewed by Dr. Jorge Silveyra, Dr. Jeffrey Pfaffmann, and Dr. Sofia Serrano prior to submission
 - Result: ICLR Workshop Paper “Faithful to What?” (arXiv:2506.12176).
- **EXCEL Scholar—Building Energy & Generative AI** June 2025 - Present
Department of Computer Science, Lafayette College Easton, PA
 - Advisor: Dr. Jorge Silveyra (Lafayette College)
 - Collaborator: Dr. Chetan Tiwari (Georgia State University)
 - Developed neural network approach to recommend energy efficiency retrofits at neighborhood scale
 - Used EnergyPlus™ simulations and generative AI for urban building energy modeling
 - First-authored manuscript submitted to *Building Simulation* (arXiv:2509.09794)
 - Presented research at KINBERCON 2025 and Lafayette poster sessions
 - Invited to present at Georgia State University
- **ACL Manuscript Review** October 2025 - Present
Department of Computer Science, Lafayette College Easton, PA
 - Collaborated as a secondary reviewer with Dr. Sofia Serrano on peer review of ACL Rolling Review manuscripts
 - Evaluated research methodology, experimental design, and validity of scientific claims
 - Drafted preliminary reviews for each submission which were refined through discussion with Dr. Serrano

TEACHING EXPERIENCE

- **Department of Computer Science, Lafayette College** August 2024 - Present
Teaching Assistant Easton, PA
 - Assist professors with teaching duties by helping students during class meetings.
 - Praised by students as an outstanding teacher who cares and can explain complex things in ways that make sense.
 - More information on my [teaching page](#).
- **Academic Resource Hub, Lafayette College** August 2024 - Present
Mentored Study Group Leader Easton, PA
 - Hold two review sessions per week, covering course content.
 - Produce materials such as worksheets and presentations for sessions.
 - Recognized by students for creating a welcoming and effective learning environment.
 - More information on my [teaching page](#).

SKILLS

- **Programming Languages:** Python, Java, JavaScript, C
- **Machine Learning:** PyTorch, TensorFlow, NumPy
- **Other Tools & Technologies:** L^AT_EX

HONORS AND AWARDS

- **Marquis Scholarship** May 2023
Lafayette College 
 - Demonstrated intellectual curiosity in addition to superior academic achievement.
 - Recipients display the Marquis’ ideals of global citizenship, scholarly pursuits, leadership, and concern for others.
- **Dean’s List** F23, S24, F24, S25, F25
Lafayette College
 - Achieved a Semester GPA of 3.60 or higher.

VOLUNTEER EXPERIENCE

- **Audio/Visual Specialist**
Hope Alliance Church
 - Design and operate weekly worship slides and sermon visuals
 - Configure live stream and projection systems for services
 - Run slides and broadcast during live events
 - Troubleshoot technical issues quickly and independently
 - Maintain and reset the worship space after services
 - Apply skills in graphic design, live sound, and audiovisual tech
 - Serve faithfully by supporting worship through technical service
 - **Audio/Visual Specialist**
Lafayette DiscipleMakers Christian Fellowship
 - Similar service to the above with no live stream component

July 2021 - Present


- September 2023 - Present



CERTIFICATIONS

- **Deep Learning Specialization**

June 2025

ADDITIONAL INFORMATION

Languages Spoken

English (Native Tongue), French (Fluent; Professional Competency)

Interests

Music (Jazz Vocals, Trombone, Piano, Composition); Photography and Photo Editing; Videography and Video Editing